Adaptive Agency: A Human–AI Co-Governance Framework for Socially Safe Grassroots Governance in Complex Urban Environments

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Abstract

As AI-enabled systems move from passive decision support toward bounded agentic operation, public-sector deployment creates safety challenges not captured by technical robustness alone. An AI system may apply predefined rules consistently while producing outcomes that are contextually inappropriate, difficult to contest, or institutionally unclear. This paper defines social safety as a socio-technical capacity to maintain contextual appropriateness, contestability, accountability, and non-exclusion. It develops this argument through a retrospective longitudinal reconstruction and secondary qualitative analysis of an AI-enabled governance platform in T Village, Tianjin. Published interview evidence, source-attributed indicators, and documentary materials trace the transition from paper-based mechanisms in 2017 to an integrated platform in 2024–2025. The analysis identifies a Technological Governance Power Gap between standardized algorithmic rules and contextual human judgment, mediated through multidirectional translation and mutual adaptation. Reported operational indicators and a negative case involving repeated misclassification of a delivery worker show why technical consistency does not itself establish social safety. Building on prior analysis of Power–Algorithm Adaptive Survival, the paper develops the Adaptive Agency framework and a SAFER roadmap based on Dynamic Power Alignment, Context-Aware Safety Constraints, and Mutual Adaptation. The central proposition is that safer agentic governance depends on supervised autonomy rather than autonomy maximization.

1 Introduction

Artificial intelligence is moving beyond passive prediction and content generation toward systems that can perceive changing conditions, maintain operational states, select actions, invoke tools, and execute tasks with limited supervision. Agent research increasingly emphasizes planning, tool use, environmental interaction, and extended action sequences, while also recognizing that benchmark performance alone does not establish real-world reliability or usefulness [Wang et al., 2024; Kapoor et al., 2025].

The term agentic AI nevertheless encompasses systems with substantially different levels of agency and autonomy. This paper adopts a functional and bounded definition. An AI-enabled system exhibits bounded agentic characteristics when it can continuously receive environmental information, classify situations, trigger predefined responses, and initiate limited actions without case-by-case human instruction, while remaining constrained by institutional rules, technical boundaries, and human oversight.

The T Village platform examined here is not a general-purpose autonomous agent. It is an AI-enabled socio-technical governance system that combines ordinary digital infrastructure with selected functions including sensing, automated classification, behavioral-scoring assistance, alert generation, and rule-triggered intervention. These functions can structure administrative attention and initiate bounded responses, but they do not independently define governance objectives or assume institutional responsibility.

Greater operational capability creates safety challenges that cannot be reduced to model accuracy. Established AI safety research identifies negative side effects, reward misspecification, distributional shift, unsafe exploration, and control as central problems [Amodei et al., 2016; Hadfield-Menell et al., 2017]. In public governance, however, an additional problem arises when a technically consistent classification is embedded in a socially heterogeneous setting. A person may be correctly classified as “not registered” but wrongly treated as a security threat; a scoring rule may be consistently applied yet disadvantage people whose work or living patterns do not fit standardized categories.

Such failures arise from interactions among technical classifications, institutional authority, local knowledge, affected populations, and opportunities for correction. They are therefore socio-technical rather than exclusively computational problems [Selbst et al., 2019]. This paper uses social safety to describe the capacity of a governance arrangement to maintain contextual appropriateness in non-standard situations, procedural contestability for affected people, identifiable human and institutional responsibility, and responsiveness to exclusion, error, and changing public needs.

Public governance is a critical setting for examining these issues because algorithmic systems may mediate access, direct administrative attention, assign behavioral scores, generate warnings, and influence how individuals are treated. Nor does the presence of a nominal human reviewer guarantee meaningful oversight. Human decision-makers may over-rely on system outputs, misunderstand limitations, selectively follow recommendations, or lack sufficient authority and information to correct errors [Green and Chen, 2019; Green, 2022].

These tensions are especially visible in Chinese urban villages: former rural settlements embedded within expanding metropolitan areas and characterized by high mobility, mixed residence, informal social relations, heterogeneous service needs, and limited administrative resources. T Village in Tianjin contains more than 10,000 residents, approximately 80% of whom are migrants. Its platform supported waste-sorting management, resident reporting, and administrative coordination, while also producing a socially significant failure: an access-recognition system repeatedly classified a delivery worker as high risk and publicly broadcast warnings despite his legitimate occupational need to enter residential buildings [Yang et al., 2026].

1.1 Relationship to Prior Work

This study builds on but extends the published analysis of T Village. Prior research introduced Power–Algorithm Adaptive Survival to explain how technological governance power and frontline administrative authority interact during smart-governance transformation [Yang et al., 2026]. The present paper reframes that mechanism as an agentic AI safety problem, develops the broader Adaptive Agency framework, and translates it into a SAFER roadmap for bounded autonomy, contestability, institutional oversight, and continuous evaluation.

The analysis addresses three questions: How does AI-enabled governance redistribute authority and discretion between technical systems and frontline actors? How are mismatches between standardized classifications and contextual social meaning translated and corrected? What design principles follow for socially safe agentic AI? The paper contributes a social-safety perspective, a mechanism-based account of Adaptive Agency, and a risk-tiered roadmap for supervised autonomy.

2 Related Work

2.1 Agentic AI Safety and the Missing Social Layer

AI safety research examines how intelligent systems can remain reliable and aligned under uncertainty. Early formulations emphasized negative side effects, reward hacking, unsafe exploration, distributional shift, and scalable supervision [Amodei et al., 2016]. Inverse Reward Design further showed that an observed objective should not be treated as a complete specification of human intent when systems encounter environments different from those assumed by their designers [Hadfield-Menell et al., 2017]. These concerns become more consequential as agents operate across extended action sequences and interact with external tools [Wang et al., 2024].

Cross-sector frameworks broaden safety from technical performance to lifecycle risk management. The NIST AI RMF organizes organizational practice around Govern, Map, Measure, and Manage [NIST, 2023]. The EU AI Act requires effective human oversight for high-risk systems and emphasizes the capacity to interpret, disregard, override, or interrupt system outputs [European Union, 2024]. These frameworks specify important obligations, but they offer less explanation of how local knowledge, institutional authority, public participation, and algorithmic operation interact over time.

2.2 Human Oversight, Human–AI Teaming, and Contestability

Human-centered AI research provides guidance on communicating system capabilities, representing uncertainty, supporting correction, recovering from errors, and adapting over time [Amershi et al., 2019]. Human–AI teaming research also examines reliance, explanations, confidence, task allocation, and complementary performance. Yet human presence alone does not ensure effective oversight. Algorithm-in-the-loop decision-making may fail because people cannot assess model validity, use recommendations inconsistently, or lack the power to intervene [Green and Chen, 2019]. Formal oversight requirements may consequently legitimize problematic systems without establishing effective institutional control [Green, 2022].

2.3 Public-Sector Algorithms and Frontline Discretion

Street-Level Bureaucracy explains why public-sector AI cannot be treated as ordinary workflow automation. Frontline officials exercise discretion under limited resources, ambiguous rules, and diverse citizen circumstances [Lipsky, 1980]. Digitalization redistributes that discretion by embedding policy interpretations in databases, scoring systems, classifications, and automated workflows.

Street-level algorithms occupy the gap between policy and decisions about people. Unlike human officials, who may revise their interpretation while encountering a novel case, algorithmic systems generally continue applying an existing boundary until feedback, redesign, or external review occurs [Alkhatib and Bernstein, 2019]. Empirical research similarly shows that AI-supported casework reconfigures rather than eliminates frontline discretion, shifting work toward data verification, interpretation, and coordination [Flügge et al., 2021].

3 Analytical Framework: Governance Power Gap, Contextual Translation, and Dual-Power Adaptation

3.1 Integrating Actor–Network Theory and Street-Level Bureaucracy

This study conceptualizes human–AI co-governance as a dynamic socio-technical arrangement in which technical systems, frontline administrators, residents, institutional rules, and material infrastructures jointly shape governance action. Actor–Network Theory (ANT) provides a relational account of agency and translation, while Street-Level Bureaucracy explains the institutionally authorized discretion of frontline actors [Callon, 1986; Latour, 2005; Lipsky, 1980].

Algorithmic agency here does not imply human-like intentionality or independent legal personhood. It denotes the capacity of a technical system to alter what other actors can see and do by classifying a person, calculating a score, generating a warning, prioritizing an incident, or triggering a predefined response. Algorithmic power similarly denotes operational influence rather than sovereign authority: technical classifications and workflows can structure administrative attention and affect people's treatment, while formal responsibility remains human and institutional.

3.2 The Technological Governance Power Gap

The Technological Governance Power Gap is the tension that emerges when the standardizing capacity of an AI-enabled system encounters a governance environment whose relevant meanings cannot be fully represented through predefined categories and rules. The technical side is algorithmic rigidity: systems require formal inputs, defined categories, repeatable procedures, and operational thresholds. The social side is governance elasticity: frontline action involves incomplete information, changing circumstances, interpersonal relationships, informal norms, and legitimate exceptions.

3.3 Multidirectional Contextual Translation

Contextual Translation operationalizes ANT's concept of translation for AI-mediated governance [Callon, 1986]. Three directions are analytically distinct. Human-to-system translation converts governance objectives, rules, and local experience into categories and workflows. System-to-human translation occurs when frontline actors interpret outputs through field verification and local knowledge before they become consequential actions. Society-to-platform and institutional translation converts residents' lived experience into feedback capable of changing governance rules, organizational practices, or platform functions.

3.4 Dual-Power Adaptation and Supervised Autonomy

Dual-Power Adaptation refers to the provisional and revisable allocation of operational authority between algorithmic and administrative actors. Algorithmic components contribute scale, consistency, persistence, and rapid information processing; human actors contribute contextual interpretation, authorized discretion, ethical judgment, and accountability. When selective delegation, retained intervention authority, and institutionalized revision operate together, the resulting configuration is Supervised Autonomy: bounded technical action within domains that remain risk-sensitive, contestable, traceable, interruptible, and open to institutional revision.

4 Research Methodology

This study uses a retrospective longitudinal case reconstruction and secondary qualitative analysis to examine how an AI-enabled governance platform became embedded in T Village between 2017 and 2025. It does not claim continuous direct observation. The reconstruction synthesizes previously published field evidence, source-attributed operational indicators, policy and institutional documents, and publicly available accounts of platform development [Yang et al., 2026].

T Village was selected as an information-rich and theoretically revealing case because it combines a socially complex urban-village environment, a documented transition from paper-based governance to an integrated platform, and evidence of both reported operational improvements and socially significant friction. The unit of analysis is the socio-technical arrangement comprising technical components, frontline administrators, residents, institutional rules, organizational roles, and feedback procedures.

The analysis proceeded through chronology reconstruction, source labeling, descriptive coding, pattern coding, and negative-case review. Evidence was organized around authority allocation, contextual translation, exception handling, platform revision, and accountability. The delivery-worker incident was retained as a negative case to test the limits of technically consistent rule application rather than treating reported performance indicators as sufficient evidence of social safety.

The present study did not access original interview recordings, full transcripts, field notebooks, or individual-level platform logs. Published interview excerpts and source-attributed indicators are therefore treated as secondary evidence rather than independently audited primary data. The analysis seeks mechanism-oriented analytical generalization; it does not estimate causal effects or directly measure trust, fairness, legitimacy, or the four social-safety dimensions.

5 Findings: Evidence of Adaptive Human–AI Co-governance

5.1 AI Deployment Creates a Governance Power Gap Rather Than Simple Automation

The T Village platform did not simply replace human decision-making. It created a structural tension between standardized rule execution and context-dependent judgment. In waste sorting, a "123" procedure generated a reminder after the first identified violation, a warning after the second, and a point deduction after the third. The reported participation rate exceeded 90%, and the reported correct waste-sorting rate increased from 50% to 80%. These indicators describe the operation of a standardized routine but do not independently establish broader social-safety outcomes [Yang et al., 2026].

The access-recognition incident exposed the other side of the power gap. The system consistently applied the mapping from an unregistered entrant to a high-risk warning, but the mapping was contextually invalid for a delivery worker with a legitimate occupational reason to enter. The failure was therefore not simply inconsistent classification; it was the inability of a formal category to represent the social meaning of routine mobility [Yang et al., 2026].

5.2 Social Safety Emerges Through Multidirectional Translation

AI integration depended on translation among administrators, technical systems, and residents. Situational translation adapted platform functions to local administrative needs instead of reproducing a standardized model. Code-based translation converted governance objectives into operational rules and graduated responses. Participatory translation incorporated resident reports and suggestions into organizational follow-up and platform adjustment. In 2024, the platform recorded 256 resident suggestions with a reported completion rate of 96%, indicating an institutional channel through which lived experience could enter the governance process [Yang et al., 2026].

5.3 Safe Autonomy Requires Mutual Adaptation Rather Than Algorithmic Replacement

The evidence supports supervised autonomy rather than replacement. Platform functions shifted selected routine monitoring and information-processing tasks toward digital coordination, while human actors retained field verification, exception handling, interpretation, and responsibility. T Village employed ten grid workers, approximately one for every 500 households, who used platform information to identify issues and coordinate responses. Their role was not removed; it was reconfigured toward reviewing outputs and resolving context-dependent cases [Yang et al., 2026].

Adaptation also occurred at the organizational level. Platform modules and governance rules were revised as needs changed, and resident feedback was incorporated through human decision processes. The evidence does not show autonomous model learning. It shows institutionalized revision of platform functions, workflows, and governance practices—a distinction central to the Adaptive Agency framework [Yang et al., 2026].

6 Mechanism: From Power–Algorithm Adaptive Survival to the Adaptive Agency Framework

6.1 Power–Algorithm Adaptive Survival

The central challenge of deploying agentic AI in complex social environments concerns how authority, discretion, and responsibility are redistributed when technical systems become embedded in institutional settings. Power–Algorithm Adaptive Survival is the socio-technical mechanism through which algorithmic and administrative authority continually adjust their operational boundaries to preserve governance capacity while maintaining contextual appropriateness and institutional accountability.

6.2 Mechanisms of Adaptive Agency

Three mechanisms constitute this process. Dynamic Power Alignment replaces a simple automation logic with risk-sensitive authority allocation: standardized, high-frequency functions may be delegated, while ambiguous or consequential decisions remain subject to meaningful human authority. Contextual Translation bridges formal rules and social meaning through human-to-system, system-to-human, and society-to-institutional translation. Mutual Adaptation moves governance from static alignment toward co-evolutionary adjustment as platform functions, human practices, and institutional rules are revised together.

Adaptive Agency is the theoretical configuration produced by these mechanisms. Agency is distributed across algorithmic capability, human contextual judgment, and institutional capacity for review and revision; it is not a property of the model alone. Its governance expression is Supervised Autonomy, in which technical components can act within defined boundaries while remaining contestable, traceable, interruptible, and accountable. Figure 1 summarizes this pathway and identifies contextual appropriateness, contestability, accountability, and non-exclusion as evaluation dimensions rather than empirically established outcomes.

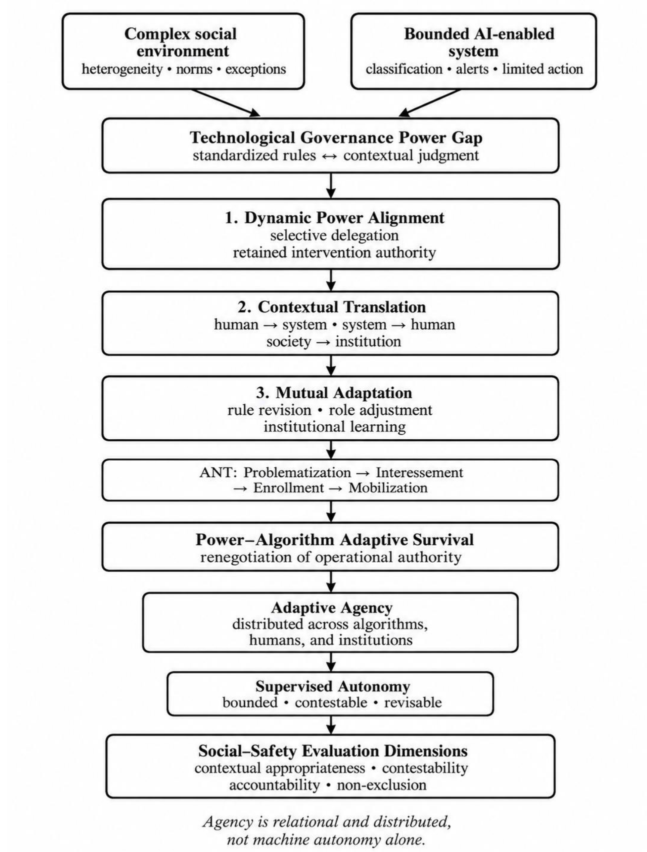

Figure 1: The Adaptive Agency Framework: From the Technological Governance Power Gap to Supervised Autonomy

7 SAFER Roadmap: Designing a Safer Agentic Ecosystem

Based on the Adaptive Agency framework, the SAFER roadmap translates the empirical mechanism into a socio-technical design architecture for bounded agentic AI. It is organized around Dynamic Power Alignment, Context-Aware Safety Constraints, and Mutual Adaptation and Institutionalized Co-evolution.

7.1 Principle 1: Dynamic Power Alignment

AI systems should augment human decision-making rather than eliminate human agency. Authority should be allocated according to task risk, uncertainty, reversibility, and potential social impact. Low-risk routine tasks may use bounded automated execution with logging and periodic audit. Medium-risk tasks may use human-on-the-loop supervision, with review of abnormal or uncertain cases. High-risk decisions involving access, sanctions, rights, dignity, or vulnerable groups should require human-in-the-loop approval.

7.2 Principle 2: Context-Aware Safety Constraints

Agentic systems require exception-aware operation, uncertainty escalation, human verification, explanation interfaces, appeal channels, correction procedures, and auditable records. Evaluation should extend beyond accuracy and latency to include false-positive patterns, intervention frequency, complaint and correction processes, contextual appropriateness, and exclusion risks. These safeguards make technical action reviewable without assuming that every social exception can be encoded in advance.

7.3 Principle 3: Mutual Adaptation and Institutionalized Co-evolution

Safety cannot be secured solely at deployment. Organizations should establish formal responsibility for impact monitoring, complaint review, coordination with technical providers, rule revision, and documentation of system updates. Adaptation refers to the capacity of the socio-technical arrangement to revise models, thresholds, workflows, and institutional practices; it should not be conflated with uncontrolled autonomous learning.

7.4 Staged Implementation Roadmap

In the near term, deployments should prioritize activity logging, risk-tiered authority, exception escalation, and accessible appeal channels. In the medium term, organizations should establish accountable governance roles, shared social-safety indicators, version tracking, and regular review cycles. In the longer term, cross-sector standards should support comparative evaluation, interoperability of audit evidence, and institutional learning across deployment contexts. The sequence moves from bounded control, to organizational capability, to ecosystem-level assurance.

8 Conclusion

As AI systems move toward bounded agentic operation, safety can no longer be treated solely as a property of an isolated model. The T Village case shows how technically consistent rules can become socially inappropriate when formal categories encounter heterogeneous mobility, informal norms, and institutional complexity. The paper therefore conceptualizes social safety as a property of an evolving socio-technical arrangement and explains it through the Power–Algorithm Adaptive Survival mechanism, Adaptive Agency, and Supervised Autonomy. Its central implication is that safer agentic AI depends on the calibrated distribution of authority among algorithms, humans, and institutions rather than on autonomy maximization.

The study has clear limitations. It is a single-case, secondary analysis and supports analytical rather than statistical generalization. The operational indicators were reported by the source study and were not independently audited; the available materials do not provide systematic measures of override frequency, false positives and negatives, appeal outcomes, correction latency, demographic disparity, or perceived fairness. Future research should test the framework across governance domains and levels of autonomy, combine direct fieldwork with system logs and longitudinal indicators, and examine whether specific oversight and contestability mechanisms measurably reduce social-safety risks.

Ethical Statement

This study conducts secondary analysis of published, role-anonymized case materials and aggregated operational indicators. It did not recruit new participants or access identifiable individual-level records. The analysis nevertheless recognizes ethical risks associated with surveillance, behavioral scoring, public warning, misclassification, unequal access to contestation, and the treatment of vulnerable groups.

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